

UNIVERSIDAD DISTRITAL FRANCISCO JOSE DE CALDAS
ENGINEERING SCHOOL
COMPUTER ENGINEERING

Computer Networking 1 (020-82)

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20182020100

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Workshop 2

1. Analysis of Decisions Made for Network Construction

For the network construction in Packet Tracer, several strategic decisions were made to ensure efficient communication and proper segmentation of services:

Network Infrastructure:

A basic topology was established with a backend server in Python (BackendPython), a backend server in JavaScript (BackendJavaScript), and a frontend server for user interaction. Static IP addresses were assigned to each server within the 193.168.100.0/24 subnet with the following distribution:

1. BackendPython: 193.168.100.201
2. BackendJavaScript: 193.168.100.202
3. Default Gateway: 193.168.100.1

Service Configuration:

An HTTP server was configured on each backend to respond to requests on the "/healthcheck" route, allowing validation of each service's operation.

The frontend was programmed to send requests to the backend servers using JavaScript fetch requests.

Network Segmentation via VLAN:

To enhance security and traffic management, a VLAN was implemented to restrict access to the BackendPython and BackendJavaScript servers exclusively to the frontend server. The switch interfaces were configured to allow traffic only between the servers within the VLAN, blocking access from devices like StudentLaptop and WorkerPC.

Testing and Validation:

Connectivity tests were performed using ping to verify communication between servers. The web page hosted on the frontend server was accessed from client devices to validate the proper functionality of backend services.

2. Analysis of Validation Using Packet Tracer Simulation Mode

The Simulation Mode in Packet Tracer allowed observation of how data packets travel through the different layers of the OSI Model:

1. Layer 1 (Physical):

The correct connection of devices was verified using appropriate cables (crossover cables between switches and end devices, and straight-through cables for standard connections).

2. Layer 2 (Data Link):

Ethernet frames encapsulated with correct MAC addresses were observed, ensuring communication between devices within the same VLAN.

3. Layer 3 (Network):

The proper routing of IP packets through the Default Gateway (193.168.100.1) was confirmed, enabling connection between servers and the frontend.

4. Layer 4 (Transport):

HTTP requests were validated to be correctly sent using the TCP protocol on port 80.

5. Layers 5, 6, and 7 (Session, Presentation, and Application):

HTTP requests and JSON responses from backend services were analyzed, ensuring data is transmitted and presented correctly to the user.

3. Conclusions

1. The simulation confirmed the correct flow of information through the OSI model, ensuring backend servers properly respond to frontend requests.
2. Implementing the VLAN enhanced network security by restricting access to backend servers.
3. The network structure meets the workshop requirements, providing a functional and secure environment for web services.

Screenshots

